

Simulating Just-in-Time Adaptive Interventions: A Config-Driven, LLM-Bootstrapped Framework

William Dennis

University of Bristol & National University of Singapore

Physical inactivity is a leading modifiable health risk [1]; just-in-time adaptive interventions (JITAI) deliver personalised prompts through wearables, but trials cannot probe counterfactuals, and patient transition data is rarely public [2]. We reduce policy design to a config-driven pipeline for two simulators: LLM-bootstrapped patient models, evaluated against an optimal dynamic-programming bound and bandit baselines; the optimal policy is essentially idle on the Phase-1 tables.

1. Motivation and approach

We formalise the intervention as a Markov decision process $\mathcal{M} = (\mathcal{S}, \mathcal{A}, \mathcal{P}, \mathcal{R})$ and build a framework in which *everything* is configuration: state spaces, action sets, transition models, rewards, and agents are declared in YAML, so experiments differ by config file alone (configs and tables in the repository) [3, 4].

2. LLM-bootstrapped transitions

The transition function $P(s' | s, a)$ is the hardest component of any patient model: it must encode how context, burden, and a delivered suggestion change behaviour, and it is rarely available as data. Rather than hand-curating tables or waiting for trials, we prompt a large language model (LLM, DeepSeek V4 Flash, with a GLM 5.2 comparison) to sample next-state distributions given a persona description, current state, and candidate action — a text world model [5]. Unlike data-driven simulators [2], which require logged trial data, bootstrapping needs no logged data — though it still requires hand-authored personas, states, and rewards in config. Park et al. [6] report 85% accuracy on General Social Survey replication for their agents; we do not claim our personas replicate population behaviour — GSS was not measured on the bootstrap. Six full-factorial sweeps of 22,320 prompts each (133,920 prompts, 10 samples per outcome) produced 31 JSON transition tables for DeepSeek-only \$24.52 (plus \$25.45 for GLM 5.2). Personas produced distinct sampled transition distributions (qualitative), and the config’s Bayesian mechanism models burden as lowering future activity probability [2, 7].

3. Validation

On the Phase-1 persona tables, a HeartSteps-V2-style UCB (upper confidence bound) bandit reaches 47–95.8% of the optimal dynamic-programming bound across personas (50 seeds; 95% confidence intervals ± 0.6 –9.5pp); the idle policy alone reaches 96.8–100.1%, the values above 100% being Monte Carlo noise (± 3 –5), because the bound is essentially the idle policy — interventions underperform idle on these tables. All baselines are internal to the bootstrapped tables (mock-on-mock; no ground truth; burden sensitivity untested). Eight literature-review reports document this background; the one quantitative comparison (a V2-style agent at 46.7% vs the paper’s 78.4%, on different data) did not reproduce — the V1 generative model is non-public.

4. PEARL-matched extension (Phase 2)

In the final two weeks we compared the simulator against the PEARL randomised controlled trial (RCT) [8], whose reinforcement-learning (RL) arm gained +295.9 steps over control ($p=0.0002$): we used 6 COM-B themes (Capability–Opportunity–Motivation–Behaviour) $\times$ 2 delivery times [9], plus idle — with an ϵ -greedy bandit and a COM-B barrier-weighted fixed arm under a four-tier evaluation constitution [10–12]. The 14,040-prompt bootstrap (108 states $\times$ 13 actions $\times$ 10 samples) passed 11 of 17 checks; the six failures (inflated effect size $d=1,525$ vs 218–296; fixed $>$ RL inversion; no attenuation; no burden saturation; inverted weekend; degenerate control t-test) trace to an idle-vs-action structure; a reward-penalty probe restores the RL-vs-fixed ordering only under the penalised reward, and the unpenalised four-arm benchmark (reward) still mirrors the inversion: fixed $49.2 \pm 15.7 >$ RL $47.5 \pm 18.1 >$ random $45.6 >$ control 10.9 , within noise — calibration metrics, not trial outcomes.

5. Conclusion

For these two simulators, policy design reduces to a cheap, config-driven pipeline, albeit mock-on-mock only, with a documented map of where the simulation diverges from published trials.

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